No Randomized Speedup for Weakly Convex Optimization

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Abstract

For globally Lipschitz weakly convex optimization, the best known methods in the local value-and-subdifferential model need order ε^{-4} queries to make the Moreau-envelope gradient at most ε. A recent lower bound proves that this rate is unavoidable for deterministic algorithms, even when every query returns the full subdifferential set. Whether randomization can improve the rate was left open.

We prove that it cannot. Every randomized algorithm requires

$$\Omega\left(\frac{G^2 \min\{\rho\Delta, G^2\}}{\varepsilon^4}\right)$$

queries on the class of G-Lipschitz, ρ-weakly convex functions with initial gap at most Δ. The oracle returns both the value and the entire subdifferential. The hard instances live in a polynomial dimension. When $\Delta \leq G^2/\rho$, the lower bound matches the known randomized upper bound in all four problem parameters.

Two obstacles distinguish the result from a standard random rotation. A full subdifferential exposes any exact dependence on an unrevealed direction, and unbounded queries defeat concentration. We give the diagonal chain a smooth dead zone that yields exact local independence. We then combine a smooth soft projection with a linear-growth radial penalty. This wrapper is globally Lipschitz, preserves weak convexity, hides unbounded queries, and introduces no spurious Moreau-stationary point. A conditional-Haar argument completes the randomized lower bound.

1 Introduction

A function $f : \mathbb{R}^d \to \mathbb{R}$ is ρ-weakly convex if $f + \frac{\rho}{2}\|\cdot\|^2$ is convex. This class contains convex-composite, robust-estimation, and phase-retrieval objectives while allowing nonsmooth and nonconvex geometry. Its standard stationarity measure is the gradient of the Moreau envelope

$$f_{1/(2\rho)}(x) = \min_z \{f(z) + \rho\|x - z\|^2\}.$$

Subgradient and model-based methods find a point with $\mathbb{E}\|\nabla f_{1/(2\rho)}(x)\| \leq \varepsilon$ using $O(\rho G^2 \Delta/\varepsilon^4)$ queries in the small-gap regime (Davis and Drusvyatskiy, 2018, 2019). The fourth power contrasts with the second-power rate of smooth nonconvex optimization. Earlier model-based work established asymptotic convergence without a rate (Duchi and Ruan, 2018). More recent deterministic work gives fourth-power guarantees for related Goldstein targets and for Moreau stationarity through bundle subproblems (Kong and Lewis, 2024; Liao and Zheng, 2025).

Li and Pan (2026) recently showed that this contrast is intrinsic for deterministic algorithms. Their oracle is unusually strong. At x, it returns $(f(x), \partial f(x))$, including the entire subdifferential set rather than a selected subgradient. Their construction couples tilted max-affine blocks into a constant-width diagonal zero-chain. A finite-horizon resisting rotation then handles arbitrary deterministic queries.

That rotation can depend on a deterministic trajectory. It does not apply to a randomized method. Drawing a rotation in advance creates two leaks. First, a small positive projection onto an unrevealed direction changes the smooth relay and appears in the full subdifferential. Second, a randomized algorithm can query at arbitrarily large norm, amplifying a random projection until the hidden direction becomes visible. The latter issue also appears in smooth nonconvex lower bounds, where it is handled by a soft projection and a quadratic penalty (Carmon et al., 2020). A quadratic is not available here because the objective must be globally Lipschitz.

We resolve both leaks. A shifted smoothstep makes the diagonal chain locally constant on a neighborhood of every unrevealed relay coordinate. The tilted facets already have strict intercept gaps. At radius $\tau = 1/(64N^2)$, these two facts give exact local independence from every direction more than one filtration layer ahead. This exact statement matters: an approximation in value or gradient would not control an oracle that returns the full subdifferential set.

For unbounded queries, we introduce the wrapper

$$H_U(x) = \frac{1}{4}\left[\bar{f}\big(U^\top s_S(x)\big) + 3\big(\sqrt{B^2 + \|x\|^2} - B\big)\right], \tag{1}$$

where $s_S(x) = x/\sqrt{1 + \|x\|^2/S^2}$ and U has orthonormal columns. The first term only sees a point of norm at most S. The second has bounded slope and prevents stationarity at infinity. Their interaction is the main technical issue. We prove that the radial term cannot cancel the chain subgradient at a proximal point. Every positive coordinate on the support of that subgradient is $O(1/N)$, while the chain norm is $\Omega(N^{-1/2})$.

Contributions. Our main theorem closes the randomized local-oracle complexity in the small-gap regime $\Delta \le G^2/\rho$. It matches both the deterministic lower bound and the known randomized upper bound, including the dependence on G, ρ, Δ, and ε. The proof also gives a reusable globally Lipschitz replacement for the quadratic soft-projection wrapper in randomized stationarity lower bounds. Finally, we formulate a robust full-set chain condition and show that a width-six diagonal filtration is compatible with the conditional-Haar argument of Carmon et al. (2020).

Related lower bounds. Random rotations are classical in randomized convex lower bounds (Woodworth and Srebro, 2017; Diakonikolas and Guzmán, 2020). The robust zero-chain and soft-projection framework of Carmon et al. (2020) treats smooth functions and local derivative oracles. Kornowski and Shamir (2022) prove a different impossibility result for finding a point near one with a small Clarke subgradient. Their class is unrestricted Lipschitz nonconvex functions, and their target is not Moreau stationarity. The closest result is Li and Pan (2026), which proves the deterministic version of our rate and explicitly leaves randomized full-set algorithms open. We retain their tilted-block architecture, but both robustness and the Lipschitz wrapper are new.

The upper-bound literature uses several oracle models. Stochastic subgradient and model-based schemes control the Moreau-envelope gradient (Davis and Drusvyatskiy, 2018, 2019), while proximally guided methods control a nearby stationarity certificate (Davis and Grimmer, 2019). The deterministic method of Kong and Lewis (2024) uses a directional subgradient oracle. Liao and Zheng (2025) obtain the same fourth-power Moreau rate with function values and subgradients through convex bundle subproblems. Our lower bound grants the algorithm the full set $\partial f(x)$, so it covers these first-order information models when specialized to the same class and target.

Structured smoothing can be faster under stronger access. The framework of Deng and Gao (2026) assumes that a prescribed smooth approximation is computationally tractable and that its gradients can be queried. Moreau smoothing then requires solving a proximal optimization problem, while Nesterov smoothing uses a known composite representation. The method of Najar (2026) similarly uses explicit composite structure and a linear minimization oracle. Such surrogate gradients are not determined by the local pair $(f(x), \partial f(x))$. Our theorem therefore separates these structured models from black-box local first-order access.

2 Model and Main Result

For $G, \rho, \Delta > 0$, let $\mathcal{F}(G, \rho, \Delta)$ be the union over dimensions of finite-valued, bounded-below functions $f : \mathbb{R}^d \to \mathbb{R}$ such that f is G-Lipschitz, ρ-weakly convex, and

$$f(0) - \inf_x f(x) \le \Delta.$$

The oracle is $\mathcal{O}_f(x) = (f(x), \partial f(x))$. Weakly convex functions are subdifferentially regular, so the usual limiting, Clarke, and regular subdifferentials agree (Rockafellar and Wets, 2009). Algorithms are measurable, may use internal randomness, start at zero, and may choose each query from the full previous transcript. An arbitrary transcript-measurable output is allowed.

Definition 1 (Randomized query complexity)**.** *For failure probability $\delta \in (0, 1)$, let $T^{\mathrm{rand}}_{\varepsilon,\delta}(\mathcal{F})$ be the least T for which a randomized algorithm can, for every $f \in \mathcal{F}$, make at most T oracle calls and output $\widehat{x}$ satisfying* $\Pr(\|\nabla f_{1/(2\rho)}(\widehat{x})\| \le \varepsilon) \ge 1 - \delta$.

Theorem 2 (Randomization does not help)**.** *There are numerical constants $c_0, c_1 > 0$ such that, whenever*

$$\varepsilon^2 \le c_0 \min\{G^2, \rho\Delta\},$$

we have

$$T^{\mathrm{rand}}_{\varepsilon,1/2}\big(\mathcal{F}(G, \rho, \Delta)\big) \ge c_1 \frac{G^2 \min\{\rho\Delta, G^2\}}{\varepsilon^4}.$$

The hard function can be taken in a dimension polynomial in G/ε.

When $\Delta \le G^2/\rho$, projected subgradient descent (Davis and Drusvyatskiy, 2018) and Theorem 2 give the minimax rate

$$\Theta\left(\frac{\rho G^2\Delta}{\varepsilon^4}\right). \qquad (2)$$

Indeed, the optimized bound of Davis and Drusvyatskiy (2018, Corollary 2.2), followed by Markov's inequality, gives success probability at least 1/2 at this query order. For larger gaps, Theorem 2 still gives the saturated lower bound $\Omega(G^4/\varepsilon^4)$, but we do not claim a matching gap-independent upper bound.

3 An Exactly Robust Diagonal Chain

We first construct an unscaled instance on M blocks of N coordinates. Assume $N \geq 60$ and $2 \leq M \leq N/12$. Write $z = (z^1, \ldots, z^M) \in \mathbb{R}^{MN}$. Define

$$c_j = 1 + \frac{N-j}{16N},$$
$$\phi(u) = \max\left\{0, \max_{j\in[N]}\left(c_j - \frac{N}{8}u_j\right)\right\}.$$

Let $q(t)$ equal zero for $t \leq 0$, $3t^2 - 2t^3$ for $0 < t < 1$, and one for $t \geq 1$. We insert a dead zone by setting

$$q_a(t) = q\left(\frac{t-a}{1-a}\right), \qquad a = \frac{1}{16},$$
$$\psi(u) = \frac{1}{N}\sum_{j=1}^{N} q_a\left(\frac{N}{4}u_j\right).$$

With $\psi(z^0) = 1$, define phase scores

$$\sigma_m(z) = \phi(z^m) + \psi(z^{m-1}) - \frac{5}{4}$$

and the hard function

$$\bar{f}(z) = \frac{2}{N}\left(\|[\sigma(z)]_+\|_2 - \sum_{m=1}^{M}\psi(z^m)\right). \qquad (3)$$

This is the tilted-block construction of Li and Pan (2026) with q replaced by q_a.

The analytic structure is visible from a variational representation. Let $W = \{w \in \mathbb{R}^M_+ : \|w\| \leq 1\}$ and set $w_{M+1} = 0$. The support-function identity for the positive orthant gives

$$\bar{f}(z) = \max_{w\in W} F_w(z),$$
$$F_w(z) = \frac{2}{N}\left(w_1 - \frac{5}{4}\sum_{m=1}^{M} w_m\right) + \frac{2}{N}\sum_{m=1}^{M} w_m\phi(z^m)$$
$$- \frac{2}{N}\sum_{m=1}^{M}(1-w_{m+1})\psi(z^m). \qquad (4)$$

Every F_w has weak-convexity modulus at most 64/75. Indeed, the tilted terms are convex and the negative progress terms have this common curvature bound. The pointwise maximum preserves it. The same formula gives the full max-rule subdifferential used below, including all active tilted facets.

Let $L = \lfloor 3N/16 \rfloor$ and assign coordinate (m, j) the level

$$\ell(m, j) = (m-1)L + j - 1.$$

Let V_t span the coordinates of level at most t, with $V_{-1} = \{0\}$. Every layer has at most six coordinates. Let P_t denote orthogonal projection onto V_t and set $\tau = 1/(64N^2)$.

Proposition 3 (Robust chain certificates)**.** *The function $\bar{f}$ is bounded below, one-Lipschitz, and one-weakly convex. It satisfies $\bar{f}(0) - \inf \bar{f} \leq 3M/N$. Moreover:*

1. *For any integer $t \geq -1$, if $|z_{m,j}| < \tau$ whenever $\ell(m, j) > t$, then throughout a neighborhood of z,*

$$\bar{f}(y) = \bar{f}(P_{t+1}y).$$

2. *Set $b = q_a(3/4) = 2783/3375$. If $\psi(z^M) < b$, then $R(z) := \|[\sigma(z)]_+\|_2 > 0$. Every $g \in \partial\bar{f}(z)$ is coordinatewise nonpositive and satisfies*

$$\|g\| \geq \frac{1}{4\sqrt{N}}.$$

If $g_{m,j} < 0$ and $z^m_j > 0$, then $z^m_j < 13/(2N)$.

The first statement is exact. For a partially revealed block, the first hidden facet beats the next one by at least

$$\frac{1}{16N} - \frac{N\tau}{4} > 0.$$

For an unreached block, its phase score is at most $-1/(16N) + N\tau/8 < 0$. All hidden progress coordinates lie strictly inside the dead zone. To see the complete filtration argument, let

$$k_m = \min\{N, \max\{0, t - (m-1)L + 1\}\}.$$

If $1 \leq k_m < N$, the progress term uses only coordinates through k_m, and the strict facet comparison makes $k_m + 1$ the only possibly relevant hidden facet. If $k_m = 0$ and $m \geq 2$, then $k_{m-1} \leq L$, so $\psi(z^{m-1}) \leq 3/16$ and the phase score is strictly negative. When $m = 1$ and $t = -1$, only its first facet can matter. Fully revealed blocks already lie in V_t. The strict margins persist on a neighborhood, proving local equality with the projection onto V_{t+1}.

For the second statement, $R(z) > 0$ makes the active outer weight unique. The tilted-facet and negative-progress terms in every subgradient are both coordinatewise nonpositive, so they cannot cancel. Their block norms sum to at least $1/(16N)$. Positivity of a supporting coordinate forces either an active facet above $1/4$ or an unsaturated progress derivative. Both give the $13/(2N)$ bound. Complete proofs appear in the supplement.

4 A Lipschitz Wrapper for Unbounded Queries

Let $n = MN$, let $U \in \mathbb{R}^{d\times n}$ satisfy $U^\top U = I_n$, and set

$$B = 512\sqrt{M}, \qquad S = 16B,$$
$$s_S(x) = \frac{x}{\sqrt{1 + \|x\|^2/S^2}}.$$

Define H_U by (1). Unlike the quadratic wrapper used for smooth lower bounds, its radial term has slope at most three.

Proposition 4 (Wrapper certificates)**.** *For every U, the function H_U is bounded below, one-Lipschitz, and one-weakly convex. Its initial gap is at most $3M/(4N)$. If x satisfies*

$$|\langle u_{m,j}, s_S(x)\rangle| < \tau, \qquad j > \lfloor N/2 \rfloor, \tag{5}$$

then

$$\|\nabla (H_U)_{1/2}(x)\| > \frac{1}{64\sqrt{N}}.$$

The class certificates follow from composition calculus. The map s_S is one-Lipschitz and its Jacobian is $6/S$-Lipschitz. Composing a one-Lipschitz, one-weakly convex function with this map gives weak-convexity modulus at most $1 + 6/S$. The factor $1/4$ restores modulus and Lipschitz constant below one. The radial term is convex and nonnegative.

We outline the stationarity argument. Suppose the claimed bound fails, set $z = \operatorname{prox}_{\frac{1}{2}H_U}(x) := \arg\min_y\{H_U(y) + \|y - x\|^2\}$, and write $d = 2(x - z)$. Then $d \in \partial H_U(z)$. If $\|z\| \geq B$, the radial subgradient has norm at least $3/\sqrt{2}$ before the factor $1/4$, while the hard term has norm at most one. Hence $\|z\| < B$.

Let $\alpha = (1 + \|z\|^2/S^2)^{-1/2}$ and $w = U^\top z$. For some $g \in \partial \bar{f}(\alpha w)$, projection of $4d$ onto the columns of U gives

$$U^\top(4d) = \alpha g + cw, \qquad c = \frac{3}{\sqrt{B^2 + \|z\|^2}} - \frac{\alpha^3\langle w, g\rangle}{S^2}. \tag{6}$$

Here $0 < c < (3 + 1/256)/B$. Condition (5), the proximal displacement, and the quadratic bound $q_a(t) \leq (256/75)t_+^2$ imply $\psi((\alpha w)^M) < b$. Proposition 3 applies.

Only positive entries of w can cancel the nonpositive vector g. On $\operatorname{supp}(g)$, every such entry is at most $13/(2\alpha N)$. Therefore

$$\|\alpha g + cw\| \geq \left[\frac{\alpha}{4} - \frac{(3 + 1/256)13}{1024\alpha}\right]\frac{1}{\sqrt{N}} > \frac{1}{5\sqrt{N}}.$$

After dividing by four, this contradicts $\|d\| \leq 1/(64\sqrt{N})$.

5 Random Rotation and Proof of the Main Theorem

Draw U uniformly from the Stiefel manifold of $d \times n$ orthonormal frames. Order its columns by nondecreasing filtration level, breaking ties in a fixed way. We use a direct adapted conditional-Haar fact. If v_k is measurable with respect to an independent seed and columns $u_1, \ldots, u_{k-1}$, with $\|v_k\| \leq S$, then conditional on this history every remaining column has a uniform spherical marginal in a subspace of dimension at least $d - n + 1$. The standard spherical-cap bound used by Carmon et al. (2020, Appendix B.3), followed by a union bound, gives $|\langle u_j, v_k\rangle| < \tau$ simultaneously for all $k \leq j$ when (7) holds.

Lemma 5 (Robust random rotation)**.** *Fix a randomized algorithm and $\delta \in (0,1)$. If*

$$d \geq n + 2S^2\tau^{-2}\log\frac{2n^2}{\delta}, \tag{7}$$

then, with probability at least $1-\delta$ over U and the algorithm, every query x_t before filtration level H satisfies

$$|\langle u_{m,j}, s_S(x_t)\rangle| < \tau \quad \textit{whenever } \ell(m,j) \geq t.$$

Here is the reduction to that geometric fact. Let $\pi(1), \ldots, \pi(n)$ be the column order and let $K_t = \{k : \ell(\pi(k)) = t\}$. Define a surrogate transcript on every history by replacing $\bar{f}(y)$ with $\bar{f}(P_t y)$ in the wrapped oracle response at round t. Its effective query $\widetilde{y}_t = s_S(\widetilde{x}_t)$ is measurable from the random seed and columns in levels below t. For each $k \in K_t$, set $v_k = \widetilde{y}_t$, and pad unused later entries with zero. Reordering a Haar frame preserves its law. Moreover, v_k does not use any column in its own layer, so the sequence is adapted even when a query is repeated across a width-six layer.

Apply the conditional-Haar fact to this virtual sequence. On its good event, suppose the surrogate and true transcripts agree through query t. Every coordinate of $U^\top s_S(x_t)$ at level at least t then has magnitude below τ. Proposition 3, with index $t-1$, says that $\bar{f}$ agrees locally with $\bar{f} \circ P_t$. Thus their values and entire subdifferentials agree. The exact chain rule through

$U^\top s_S$ makes the two wrapped oracle responses equal, and the next queries agree by induction. This proves the lemma. The supplement provides the full adapted-Haar statement and the measurability details.

We finish the construction. Let

$$\kappa = \frac{1}{64}, \qquad r = \min\left\{\frac{\rho\Delta}{G^2}, 1\right\},$$
$$N = \left\lfloor \frac{\kappa^2 G^2}{\varepsilon^2} \right\rfloor, \qquad M = \left\lfloor \frac{Nr}{12} \right\rfloor.$$

Under the accuracy assumption of Theorem 2, these integers are admissible. Scale the wrapper as

$$F_U(x) = \frac{G^2}{\rho} H_U\left(\frac{\rho}{G}x\right). \tag{8}$$

Then $F_U \in \mathcal{F}(G, \rho, \Delta)$ and

$$\|\nabla (F_U)_{1/(2\rho)}(x)\| = G \left\|\nabla (H_U)_{1/2}\left(\frac{\rho}{G}x\right)\right\|.$$

Set $H = (M-1)L + \lfloor N/2 \rfloor$. We have $H \geq MN/120$. Apply Lemma 5 to the induced algorithm whose wrapper query is $\rho x/G$. Index its oracle calls as $x_0, \ldots, x_{T-1}$, and append its transcript-measurable output as x_T. If $T < H$, the output round is still before level H. Lemma 5 and Proposition 4 therefore show that the output is not ε-stationary with probability at least $1-\delta$. Set $\delta = 1/4$. Averaging over U gives a fixed U on which the algorithm succeeds with probability at most $1/4$, which is strictly below the required $1/2$. Using $N = \Omega(G^2/\varepsilon^2)$ and $M = \Omega(Nr)$ proves Theorem 2.

The dimension in (7) is $O(MN^4 \log(MN))$, and hence polynomial in the inverse accuracy. We have not optimized numerical constants.

6 Discussion

Theorem 2 removes randomization as a possible explanation for the fourth-power complexity of weakly convex optimization. The obstruction persists under a full-subdifferential oracle, so it does not arise from an adversarial choice of one subgradient. In the small-gap regime, the existing upper bound makes the nonsmooth penalty in (2) minimax exact up to numerical constants.

Three qualifications are important. First, the construction uses a high but polynomial dimension. Dimension-sensitive rates remain open. Second, the oracle is local and first order. A proximal oracle can reveal global information and supports faster rates for structured subclasses. Third, the dead zone is essential for a full-set oracle. Random rotation alone gives only small hidden projections, while any nonzero progress derivative would encode an unrevealed column exactly.

The wrapper may be useful beyond this problem. Smooth lower bounds can use a quadratic radial penalty because global function Lipschitzness is absent. The linear-growth penalty in (1) retains the same information shield while respecting a global Lipschitz budget. Its noncancellation proof uses a sign-and-support certificate rather than smooth gradient geometry.

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